GLVMamba: A Global–Local Visual State-Space Model for Remote Sensing Image Segmentation

Huihui Li, *Member, IEEE*, Huajian Pan, Xiaoyong Liu, Jinchang Ren, *Senior Member, IEEE*, Zhiguo Du, *Senior Member, IEEE*, and Jingjing Cao

***Abstract*— Semantic segmentation of remote sensing images (RSIs) has significant advances with the adoption of deep neural networks, taking the advantages of convolutional neural networks (CNNs) in local feature extraction with transformers in global information modeling. However, due to the limitations of CNNs in long-range modeling capabilities and the computational complexity constraints of transformers, remote sensing (RS) semantic segmentation still faces issues such as serious holes, rough edge segmentation, and false and even missed detections caused by the light, shadow, and other factors. To address these issues, we propose a visual state-space (VSS) model called GLVMamba, which uses CNNs as the encoder and the proposed global–local VSS (GLVSS) block as the core decoder. Specifically, the GLVSS block introduces locality forward feedback and shift window mechanism to addresses the deficiency of insufficient modeling of neighboring pixel dependencies of Mamba, which enhances the integration of global and local context during feature reconstruction, boosts object perception capabilities of the model, and effectively refines edge contours. In addition, the scale-aware pyramid pooling (SCPP) module is proposed to fully merge the features from various scales and adaptively fuse and extract the distinguishing features to mitigate the holes and false detections. The GLVMamba effectively captures global–local semantic information and multiscale feature through the GLVSS block and the SCPP module, achieving efficient and accurate RS semantic segmentation. Extensive experiments on two widely used datasets have effectively demonstrated the superiority of our proposed method over the other state-of-the-art methods. The code will be available at https://github.com/Tokisakiwlp/GLVMamba**

***Index Terms*— Mamba, remote sensing (RS), scale-aware pyramid pooling (SCPP), semantic segmentation.**

Received 17 December 2024; revised 21 February 2025 and 12 April 2025; accepted 6 May 2025. Date of publication 23 May 2025; date of current version 19 June 2025. This work was supported in part by Guangdong Science and Technology Project under Grant KTP20210197 and Grant KTP20240126, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515010939, and in part by the Project of Education Department of Guangdong Province under Grant 2022KTSCX068. *(Corresponding authors: Xiaoyong Liu; Jinchang Ren.)*

Huihui Li is with the School of Computer Science, Guangdong Polytechnic Normal University, Guangzhou 510630, China, and also with Guangzhou Key Laboratory for Digital Science and Technology, Guangzhou Digital Technology Group Company Ltd., Guangzhou 510656, China (e-mail: lihh@gpnu.edu.cn).

Huajian Pan and Jingjing Cao are with the School of Computer Science, Guangdong Polytechnic Normal University, Guangzhou 510630, China (e-mail: panhuajian@gpnu.edu.cn; caojj@gpnu.edu.cn).

Xiaoyong Liu is with the School of Computer Science, Guangdong Polytechnic Normal University, Guangzhou 510630, China, and also with the School of Data Science and Artificial Intelligence, Jilin Engineering Normal University, Changchun 130052, China (e-mail: liuxy@gpnu.edu.cn).

Jinchang Ren is with the School of Computer Science, Guangdong Polytechnic Normal University, Guangzhou 510630, China, and also with the National Subsea Centre, Robert Gordon University, AB21 0BH Aberdeen, U.K. (e-mail: jinchang.ren@ieee.org).

Zhiguo Du is with the School of Mathematics and Informatics, South China Agricultural University, Guangzhou 510642, China (e-mail: bluetooth@scau.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3572127

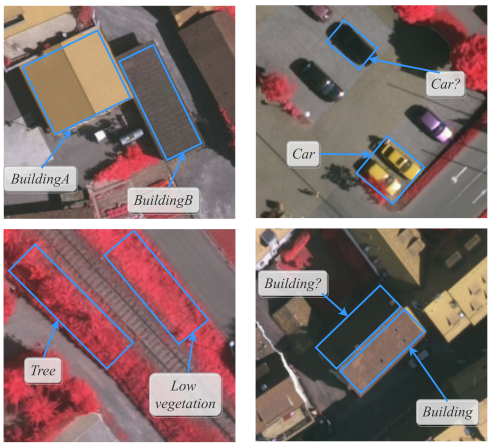
I. Introduction

ADVANCEMENTS in drone and optical imaging technology have made remote sensing images (RSIs) easily accessible. Semantic segmentation has become highly effective approach for pixel-level classification, which is a hot topic in RSI interpretation [1], [2]. Remote sensing (RS) semantic segmentation is applied in many real-world scenarios, such as urban planning [3], [4], land cover mapping [5], environmental protection [6], [7], [8], and disaster assessment [9]. In recent years, deep learning has made breakthrough progress in the field of semantic segmentation. However, due to issues such as serious holes, rough edge segmentation, false and even missed detections caused by the light and other factors, as well as high intraclass variability and high interclass similarity, which are shown in Fig. 1, RS semantic segmentation remains extremely challenging.

Semantic segmentation in RSIs based on convolutional neural networks (CNNs) has produced fruitful research results to address these challenges due to excellent feature extraction capabilities of CNNs. Fully convolutional networks (FCNs) [10], [11], [12] and SegNet [13], [14], [15] excelled in hierarchical feature learning and achieved good segmentation results. Encoder–decoder architectures such as U-Net [16], [17] [18] introduced refined upsampling and skip connections, allowing to reduce the loss of spatial information during the feature reconstruction process. DeepLabv3+ [19], [20], [21] and PSPNet [22], [23] used a multiscale pyramid strategy to capture features at various scales. However, those approaches of integrating semantic features through only convolution fail to learn long-range dependencies and spatial correlations, making them insufficient to solve the problem of misclassification caused by interclass similarity and intraclass differences.

With the advantages of transformers in global information modeling, the researchers combined CNNs with transformers. CTFuse [24] enhanced the feature representation of small-size targets and large-size ground targets by integrating attention mechanisms from both CNNs and transformers. ST-UNet [25] used the spatial interaction module and the relational aggregation module to improve the representation of occluded

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 1. Examples of the challenges in semantic segmentation of RSIs, which are taken from the Vaihingen dataset. (a) Intraclass variability and interclass similarity. (b) Affected by the shadow cast by the light, it is hard to recognize the "Car" and "Building" and the edges are blurred.

objects. TCUNet [26] combined convolution and transformer with dual encoders and used a multisource feature integration module across scales in decoding to enhance boundary and small-object segmentation. However, the lack of utilization of global–local contextual information makes the problem of objects and their boundaries blurring due to shadow occlusion not well-resolved. In addition, hybrid CNNs and transformers architectures have limitations due to CNNs have a limited receptive field, and the self-attention mechanism of transformers is computationally expensive.

Lately, the proposal of Mamba [27] provides us with new ideas. State-space models (SSMs) enable efficient long-range modeling with linear complexity, though their 1-D scanning mechanism remains suboptimal for 2-D visual data. Vision Mamba [28] leveraged bidirectional Mamba blocks for efficient visual data processing, providing faster and more memory-friendly representation learning compared with self-attention mechanisms. VMamba [29] introduced visual state-space (VSS) blocks and 2-D selective scanning (SS2D) modules, where SS2D's four scanning routes bridge 1-D ordered scanning with 2-D visual data, enhancing contextual information collection from diverse perspectives. In RSI classification, RSMamba [30] converted 2-D images into 1-D sequences and used a multipath SSMs encoder to capture long-range dependencies with excellent performance. RS^{3}Mamba [31] combined CNNs and VMamba to enhance dual-encoder features through collaborative a completion module. However, current approaches based on Mamba in RSIs are simply to replace some of the network blocks with VSS blocks containing the SS2D modules, ignoring the shortcomings of SS2D applied to RSIs, i.e., the SS2D scanning mechanism struggles to maintain the proximity between the original neighboring tokens, making it difficult to adequately extract local details and sufficiently integrate local and global contextual information.

To address the challenges in semantic segmentation of RSIs, CNN-based methods have limitations in long-range modeling capabilities, methods based on transformers are constrained by computational complexity, and Mamba-based methods fail to effectively integrate local and global contextual information. Therefore, we propose a semantic segmentation model called GLVMamba. GLVMamba consists of a CNN encoder to realize the extraction of high-resolution feature information and a decoder consisting of the improved global–local VSS (GLVSS) block. The proposed GLVSS effectively uses the SSM to capture global contextual information of linear complexity and preserves important local context. The features are further optimized by exploring the correlation between global and local contexts to improve the object blurring problem. Moreover, the scale-aware pyramid pooling (SCPP) module is proposed to capture multiscale features and adaptively select and fuse the distinguishing features to effectively mitigate the holes and false detections. The primary contributions of this article are as follows.

1) We propose a VSS model called GLVMamba, which uses the CNNs as the encoder and the proposed GLVSS block as the core decoder. GLVMamba effectively exploits powerful global context modeling capability of Mamba while enhancing the integration of local and global contexts during feature reconstruction, thereby improving segmentation performance.
2) The proposed GLVSS block introduces the locality forward feedback and a shift window mechanism, complementing the local context through local branch to address the problem of insufficient local detail extraction in the SS2D. The GLVSS block boosts the object localization capabilities of the model and effectively refines the edge contours.
3) To better extract and fuse the multiscale features, we propose the SCPP module to adaptively fuse and extract the distinguishing features to solve the problem of holes and false detections due to light and other factors.
4) Extensive experimental results on the Potsdam and Vaihingen datasets demonstrate that the proposed GLVMamba outperforms the state-of-the-art methods, thereby verifying its effectiveness and superiority.

II. Related Works

A. RS Semantic Segmentation Based on CNNs and Transformer

As CNNs gain prominence, researchers are actively delving into the potential of it in the realm of RS. FCN [10] achieved to produce more accurate classifications in shorter computation time only through convolutional and deconvolutional operations, providing a way to learn contextual features of large-scale image labeling for RSI classification. Chen et al. [32] implemented semantic segmentation to assess land use in high-resolution RS data. Their proposed SNFCN effectively eliminates the "salt-and-pepper" phenomenon and the block effect by postprocessing with a stacking strategy. F^{2}HNN [33] was a hypergraph neural network for hyperspectral image (HSI) classification that fused spatial, spectral, and multimodal features to enhance accuracy over traditional GCN

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

methods. Dang et al. [34] introduced a novel semi-supervised semantic segmentation framework that used cross-window consistency and a dynamic pseudolabel memory bank to more effectively leverage unlabeled data, achieving improved segmentation performance across diverse datasets. SPGAN-DA [35] achieved unbiased image translation from the source domain to the target domain using a semantic-preserved generative adversarial network, while reducing the feature gap between the two domains through a class distribution alignment module, demonstrating superior performance on multiple RS cross-domain segmentation benchmarks. Lu et al. [36] presented a weakly supervised semantic segmentation method for RSIs that combined self-supervised learning with pseudolabel noise reduction to improve feature representation and segmentation accuracy while reducing dependence on costly pixel-level annotations.

After U-Net [16] was proposed, it could effectively alleviate the above problems, and its encoder–decoder structure has become widely adopted. DeepUNet [37] improved the effect of sea–land segmentation using DownBlocks in contracted paths and UpBlocks in expanded paths. On the other hand, for the road segmentation problem, C-UNet [38] proposed a multiscale dense dilated convolutional UNet module for better extraction of road features, while CFAMNet [39] used the class feature attention mechanism to improve interclass correlation and further improve segmentation accuracy.

With the introduction of Vision Transformer [40], the sequence modeling capabilities of transformers were successfully adapted to image processing. Zhou et al. [41] introduced a single encoder with dual decoders based on the Swin Transformer, leveraging a global stream and a shape stream to enhance global context understanding and boundary information capture for more precise semantic segmentation of small targets in RSIs. Zhang et al. [42] introduced a hybrid transformer–CNN model with a Swin Transformer encoder and CNN-based decoder, enhanced by multiscale and edge-aware features, achieving high semantic segmentation accuracy. Chen et al. [43] combined global context modeling with detailed spatial feature extraction and multitask learning to improve the precise semantic segmentation performance of ultrahigh-resolution RSIs. Furthermore, recent technological advances have spurred exploration of large models such as SAM [44], which is trained based on the Vision Transformer architecture, and by pretraining on massive amounts of data, these models are able to learn rich features and contextual information to achieve higher segmentation accuracy on a wide range of scenarios and objects.

In our approach, we use an encoder–decoder structure, using convolutional operations and multiscale pyramids in the encoder to efficiently extract image features. On the stage of the decoder, we leverage improved Mamba to reconstruct the global–local context of the image, resulting in the attainment of precise segmentation outcomes.

B. Mamba-Based Semantic Segmentation

The utilization of CNNs and transformer-based models is prevalent in tasks such as semantic segmentation. However, CNNs are constrained by limited receptive fields, while transformers still exhibit suboptimal squared-level computational complexity. Mamba [27], as a representative of SSMs, has excelled in the domain of NLP with its linear complexity sequences modeling and has increasingly attracted research attention. VMamba [29] and Vision Mamba [28] implemented SSM in image processing by introducing a bidirectional scanning SSM module and a cross-scan module. In addition to the above areas, Mamba is also gradually applied to RS. Dong et al. [45] proposed a novel framework called dual hyperspectral Mamba (DHM), which integrated global long-range dependencies with local context information to enhance the efficiency and precision of HSI reconstruction. RSMamba [30] introduced a location-sensitive dynamic multipath activation mechanism to address the limitation that Mamba was limited to modeling causal sequences and was insensitive to spatial location, effectively accomplishing the RSI classification task. In RS change detection, CDMamba [46] combined Mamba with the convolutional network to extract local details and dynamically fused dual time-phase features through the designed AGLGF module, which guided the interaction of global and local features and extracted more discriminative change features, thus realizing more accurate change detection. Zhou et al. [47] introduced the Mamba-in-Mamba architecture for HSI classification, which combines a novel Mamba-cross-scan mechanism to improve feature aggregation and training efficiency. Liu et al. [48] leveraged ImageNet pretraining to enhance performance, outperforming CNNs, ViTs, and other Mamba-based models by a significant margin on multiple medical datasets. Sanjid et al. [49] introduced Mamba HUNet, a novel medical image segmentation architecture that combined CNNs and SSMs which achieved robust performance in tasks such as multiple sclerosis lesion segmentation on MRI scans.

In the realm of semantic segmentation, Sigma [50] was designed for multimodal semantic segmentation, leveraging Mamba to achieve global receptive fields with linear complexity. It effectively fused information from multiple modalities, outperforming traditional methods in RGB-Thermal and RGB-Depth segmentation tasks. VM-UNet [51] was a medical image segmentation model that used a U-shaped architecture built on VSS blocks, achieving effective long-range interaction modeling with linear complexity. Samba [52] replaced Mamba with multihead self-attention in ViT to capture image data information and combined it with multiple multilayer perceptrons (MLPs) to create a Samba block, which used the Samba block as an encoder and UperNet as a decoder to efficiently extract multilevel semantic information. RS^{3}Mamba [31] leveraged a dual-branch network to effectively tackle the challenge of limited local receptive fields in CNN, resulting in a notable enhancement in semantic segmentation accuracy for RSIs. SSMs (especially Mamba) provided an effective solution to the problem of excessive complexity over long-distance dependencies in various contemporary applications. RTMamba [53] effectively enhances the object perception and boundary detection in RSIs by combining the Mamba and ITP modules, which improves the segmentation accuracy of real-time RS semantic segmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

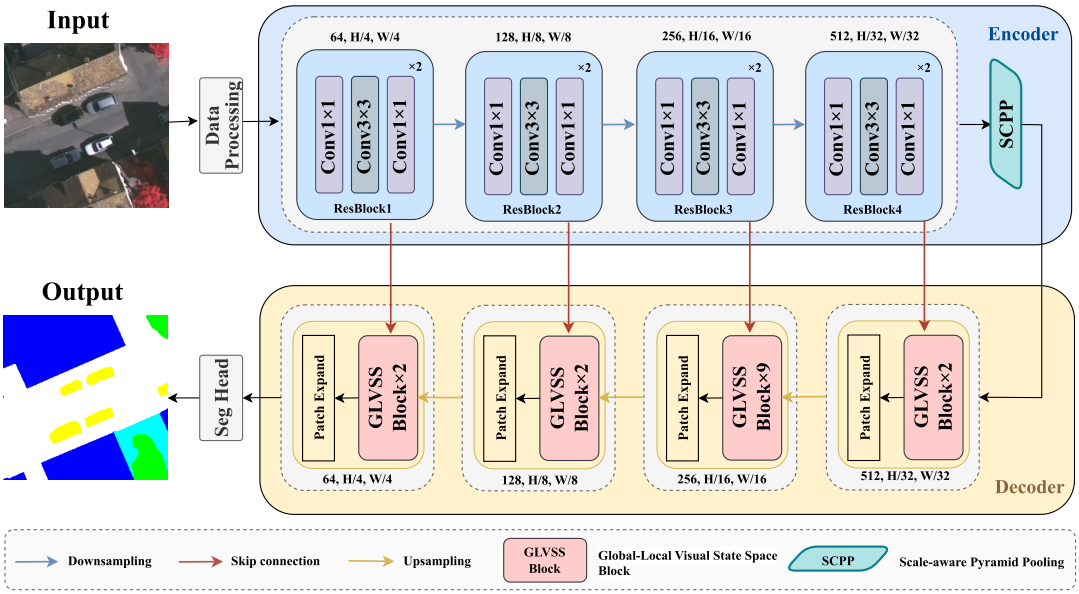


Fig. 2. Overall architecture of GLVMamba, which includes an encoder that features an SCPP module, using the ResNet as the backbone for effective feature extraction at multiscales. The decoder is constructed by a GLVSS block, which thoroughly captures both the global and local contextual information. The codec enhances the spatial information that may be missing from skip connections.

However, the above methods of using Mamba only simply replace network blocks with Mamba blocks, ignoring the significance of global and local context interaction. To solve this problem, we use the GLVSS block as a decoder, which effectively uses the global context capturing capability of Mamba while complementing the local details through local branches.

C. Multiscale Feature in RS

RSIs usually contain complex backgrounds, which may cause the distinction between targets and backgrounds to become difficult. Traditional downsampling approaches frequently fail to address this challenge effectively, prompting multiscale feature fusion to emerge as a key research focus. FPN [54] was first applied to target detection to deal with multiscale changes by fusing features at different levels. Wang et al. [55] introduced FPN into RSI segmentation by combining residual structure and UNet, which enhanced the accuracy of identifying building changes. A-FPN [56] built significant feature layers step by step using top pyramid, suppressed background interference, improved the anti-interference ability against complex backgrounds (e.g., clouds, waves, and man-made buildings), and performed well in ship detection. DeepLabv3+ [19] fused features from various scales through the ASPP module. While it improved the semantic segmentation performance, its parallel structure struggled with imprecise edge segmentation and irregularities in large-scale target segmentation within RSIs. Liu et al. [57] proposed to combine the dual attention mechanism module with ASPP, which used correlations between different channel features to strengthen the relevant class features and improve the classification, effectively eliminating the large-scale hole phenomenon of target segmentation and the problem of inaccurate edge segmentation. MFVNet [58] is a novel network designed for semantic segmentation of RSIs, addressing challenges of large field of view (FOV) processing using multiple FOVs through pyramid sampling. Ma et al. [59] presented MSPM, a comprehensive framework for HSI classification using superpixel segmentation and multiscale feature extraction.

Although the above multiscale approach solves the issue of complex background of RSIs to a certain extent, it still has shortcomings. When integrating features at different levels, it is expected that the model can choose appropriate feature details by itself. By adaptively selecting and fusing these multiscale features, the model can balance global and detail information, thereby improving segmentation accuracy. Therefore, we propose an improved multiscale fusion strategy, which makes the model better understand the complex scene in RSI that provides more accurate segmentation results.

III. Methods

In this section, we will outline the comprehensive structure of the innovative GLVMamba, emphasizing the pivotal roles of two essential modules: GLVSS block and SCPP module.

A. Overall Architecture

The overall model structure of GLVMamba is presented in Fig. 2. Our GLVMamba follows the encoder and decoder paradigm, which contains a backbone network encoder, a GLVSS-based block decoder, and a semantic segmentation header. Specifically, in the encoder stage, we leverage a lightweight CNN (ResNet18) as the backbone network for feature extraction. For input feature $X \in R^{H\times W\times 3}$, through backbone, we can effectively capture spatial information in the input data to obtain preliminary features. Furthermore, we integrate the SCPP module into the encoder to serve as a multiscale feature extractor to obtain more comprehensive and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

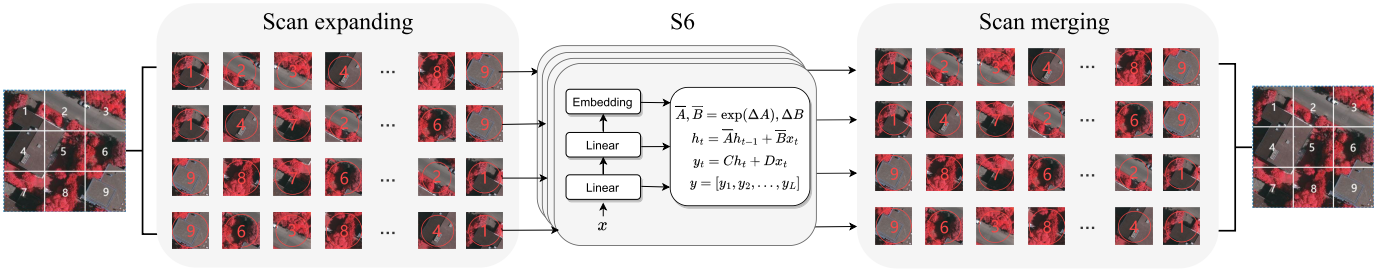


Fig. 3. SS2D process has three main steps—scan expanding, selective state-space models (S6), and scan merging: 1) scan expanding operation flattens a 2-D feature map into a 1-D vector along four directions, allowing each pixel to combine information from various angles and provide diverse viewpoints for further processing; 2) S6 processes the four 1-D vectors using the selective scanning mechanism to dynamically adjust the parameters, retaining important information while filtering out irrelevant information; and 3) scan merging operation achieves global receptive field by reintegrating 1-D sequence features extracted from different directions into a 2-D feature map of the same size as the input through an inverse operation.

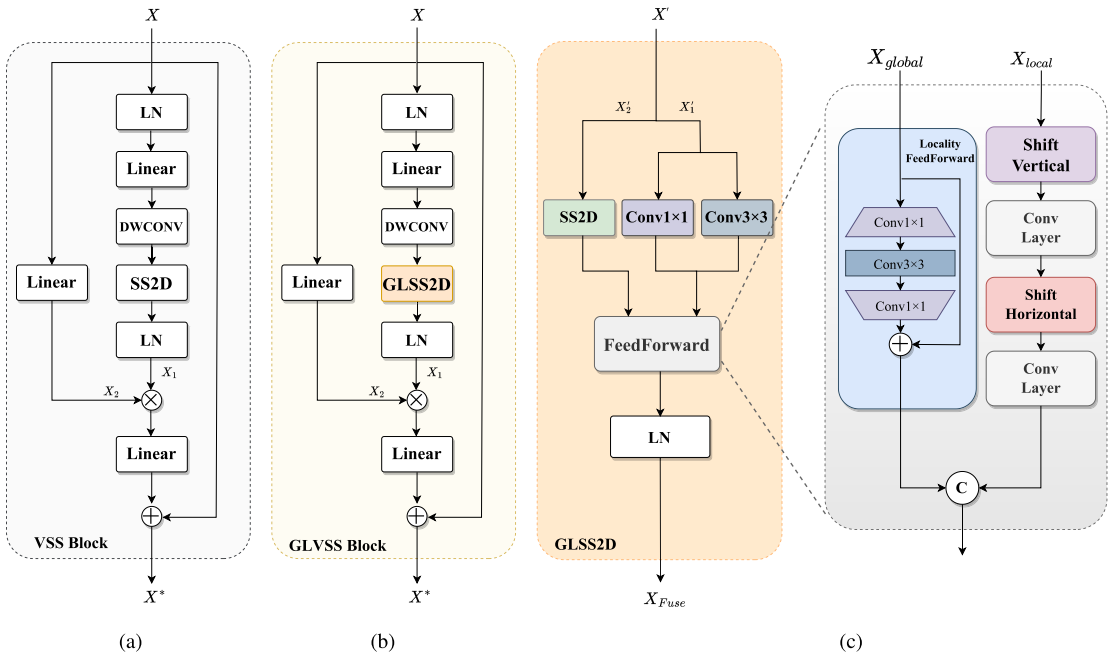


Fig. 4. Structure of the GLVSS block. (a) Overall structure of the original VSS block. (b) Overall structure of the GLVSS block. (c) Specific structure of GLSS2D. The GLSS2D includes the SS2D branch to obtain the global context and the convolutional layer to supplement the local information, and finally the feature refinement by integrating the two information through the forward propagation module.

effective feature extraction. In the decoder stage, we incorporate the GLVSS block to enhance the interactivity of global and local context information during feature reconstruction. By supplementing contextual information, the model can focus more on the connections between features. In addition, skip connections are used to feed the lower three layers of features in the encoder into the last three stages of the decoder is used to compensate for the lost spatial information. Finally, the features reconstructed by the decoder will go through a semantic segmentation head composed of bilinear interpolation and Relu activation function operations to restore the original image resolution and obtain the final prediction result image.

B. GLVSS Block

For RSIs with complex backgrounds—such as scenes containing red/brown building roofs adjacent to asphalt roads—the low visual contrast between objects poses significant challenges for accurate segmentation. In this case, local information may not provide sufficient clues for the model to distinguish between building edges and road edges. After introducing global contextual information, the model can better understand the overall scene, such as the coherent direction of roads and the overall contour shape of buildings, which helps reduce classification errors. Recent researchers have used transformers to assign global context to models, but the complex computational cost is also a problem that researchers must address. The current challenge is to reduce computational costs while maintaining accuracy.

In this regard, VMamba [29] proposed an SS2D architecture that forms sequences of images from different scanning directions and then reorganizes each sequence into a single image after constructing global dependencies through an SSM. The SS2D core structure is illustrated in Fig. 3. The architecture of the VSS block is shown in Fig. 4(a).

To expand the receptive field while maintaining computational efficiency, the scan expanding operation extends the image in four directions, creating four distinct sequences.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

Subsequently, these sequences are processed by selective SSM (S6) [27], where each element interacts with earlier scanned samples via a compressed hidden state to capture unique features. Finally, the scan merging operation recombines these sequences to the original image size. The main operation of S6 is to describe the evolution of the internal state of the system and its relationship with the input and output through a set of first-order differential equations (or difference equations). Specifically, sequences $x(t) \in R^L$ can be transformed into sequences $y(t) \in R^L$ through implicit state vectors $h(t) \in R^N$, which can be represented as

$$h_t = \hat{A}h_{t-1} + \hat{B}x_t \tag{1}$$

$$y_t = Ch_t + Dx_t. \tag{2}$$

Among them, $\hat{A} \in R^{N\times N}$, $\hat{B} \in R^N$, $C \in R^N$, and $D \in R^N$ represent the system matrices, which determine the dynamic behavior and output response of the system (where N represents the specific number of groups divided, L represents the number of images in each group, as shown in Fig. 3, where N is 4 and L is 9). More details can be found in [27].

However, there are some challenges associated with applying SS2D to RSIs. When establishing global relationships between pixels in various directions, the original spatial adjacency between neighboring pixels is disrupted, significantly increasing the distance required to capture dependencies among originally adjacent pixels (e.g., nearby car pixels may become separated [60]). This detailed connectivity is essential for analyzing RSIs. For address this issue, we design the GLVSS block to ensure that model captures the global context information while preserving the local spatial information, further realizing the feature map refinement by integrating the global–local context information. As illustrated in Fig. 4(b), the input feature $X \in R^{H\times W\times C}$ (where C denotes the number of channels of each decoder input feature, 512, 256, 128, and 64, respectively) mainly goes through two parallel branches. Within the first branch, the input feature goes through the layer normalization (LN) and linear projection operation to extend their channels to $2C$, followed by the depthwise separable convolution and Silu into the core GLSS2D module. Through this, the input feature will be enhanced with sufficient global–local information, and then finally go through the LN again. In the second, the input feature is processed only by linear projection operation and then activated by Silu. Next, the features from both the branches are aggregated by Hadamard product (elementwise product). Finally, the channel is reduced to C by a linear projection operation to produce a result matching the input features' dimensions. This process can be represented as

$$X_1 = \mathrm{LN}(\mathrm{GLSS2D}(\mathrm{SiLU}(\mathrm{DWConv}(\mathrm{Linear}(X))) \tag{3}$$

$$X_2 = \mathrm{SiLU}(\mathrm{Linear}(X)) \tag{4}$$

$$X^* = \mathrm{Linear}(X_1 \odot X_2) \tag{5}$$

where DWConv denotes the depthwise separable convolution, GLSS2D denotes the global–local SS2D, and $\odot$ denotes elementwise product.

In particular, GLSS2D also consists of two parallel branches [shown in Fig. 4(c)]. The global branch consists of the original SS2D, and the selective scanning mechanism of SS2D can effectively address the issue of the complexity of background in RSIs through the ability of long-distance dependency modeling with linear complexity and thus capturing the global context. To minimize the loss of dependency relationships between adjacent pixels, a local convolution branch is added, which contains two parallel convolutional layers with varying kernel sizes. Smaller convolution kernels are effective at capturing detailed features, while larger convolution kernels cover a broader area of information. This dual-kernel design optimally preserves adjacent pixel dependencies while strengthening the feature hierarchy.

The module's global local context capture capability is greatly improved by the local branch and the global modeling branch of SS2D. Specifically, for input feature $X' \in R^{H\times W\times 2C}$ of GLSS2D, which is divided into two distinct parts on the channel: $X'_1 \in R^{H\times W\times C}$ and $X'_2 \in R^{H\times W\times C}$. After that, the feature X'_1 undergoes local branching to capture local information to get a local feature X_{local} and another feature X'_2 extracts global context information on the image after the global branch to get global feature X_{global}.

Meanwhile, to increase the spatial correlation of local features, we incorporate the concept of window attention [61] and perform window shift operations [62] on local features along different axis directions. The window shift operation involves both the vertical and horizontal shifts. Let us take the vertical shift operation as an example. To retain essential spatial information in the input features' height dimensions, we use a zero-padding operation to prevent information loss. Next, the padding features are divided along the channel using the window size, and each group of features is shifted in height (the shift size is set to half of the window size). The shifted features are then concatenated, and the padding is removed by the narrow operation. The horizontal shift operation follows a similar process as the vertical shift operation. The convolutional layers include standard convolutional blocks, batch norm (BN), and Relu. By this method, the loss of spatial information in the process of local feature extraction can be effectively reduced. For global features, we want to model the local dependencies between nearby pixels, so we use a locality feedforward module. This module consists of two standard 1×1 convolutions and a 3×3 convolution to introduce locality into SS2D effectively. Finally, the two features are combined using concatenation, and the LN further enriches the semantic information to obtain fine-grained global and local context features. This GLSS2D processing can be represented as

$$X'_1, X'_2 \in R^{H\times W\times C} = \mathrm{Chunk}(X', 2) \tag{6}$$

$$X_{\mathrm{local}} = \mathrm{Conv}_{1\times 1}(X'_1) \oplus \mathrm{Conv}_{3\times 3}(X'_1)) \tag{7}$$

$$X_{\mathrm{global}} = \mathrm{SS2D}(X'_2) \tag{8}$$

$$X_{\mathrm{Fuse}} \in R^{H\times W\times 2C} = \mathrm{LN}(\mathrm{Cat}(\mathrm{Shift}(X_{\mathrm{local}}), \mathrm{Locality}(X_{\mathrm{global}}))). \tag{9}$$

In this context, "Chunk," "Cat," "Shift," and "Locality" refer to group operations, concatenation operations, window shift operations, and locality feedforward modules, respectively. This module gives local features the ability

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

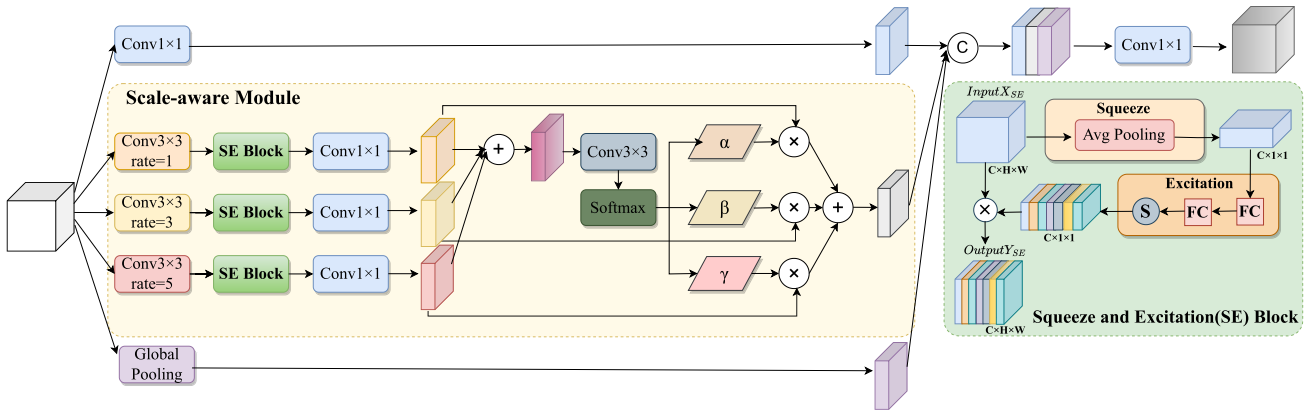


Fig. 5. Architecture of the SCPP module.

to model long-range dependencies by incorporating spatial information. To enhance global features, additional local information is also included. The combination of both the aspects further refines the feature map, leading to improved accuracy in the localization of remotely sensed objects. Furthermore, the shift operation is advantageous as it does not involve multiplication and addition, resulting in low computational complexity. Algorithm 1 presents the pseudocode for GLSS2D.

C. SCPP Module

The problem of high interclass similarity and intraclass variability in RSIs makes it difficult to distinguish object features, which can lead to misclassification and "hole" phenomena. Therefore, it is crucial to fully use features from various scales and perspectives for the semantic segmentation of RSIs. To tackle these challenges, dilated convolution aims to enhance feature representation by maximizing the receptive field and improving global context through dilated convolution with various dilated rates. However, the drawback of cross-pixel sampling in dilated convolution results in reduced spatial information, leading to ineffective perception and utilization of specific locations determined by the dilated rate.

To address this issue, inspired by Chen et al. [19], we propose a feature pyramid pooling layer with a scale-aware module, and its structure is depicted in Fig. 5. Using the scale-aware module, it can adaptively focus on important features and fully use the information of multiscale perspectives, effectively reduce the intraclass variability, mitigate the images with different noises in RSIs, and alleviate the problems of misclassification and hole phenomenon due to the large intraclass variability or the high similarity between classes.

Specifically, the SCPP module is divided into three distinct parts. As represented in Fig. 5, for the input feature X, a standard 1×1 convolution operation captures local information, while a scale-aware module extracts features at various scales, complemented by global pooling for obtaining global context. For the three inputs of the scale-aware module, the first step is to enter the three branches of parallel perceptual dilated convolution operation, where each branch contains a 3×3 dilated convolution and a squeeze-and-excitation (SE) module. Dilated convolutions with different dilated rates of sizes 1, 3, and 5 are used to enlarge the receptive field to obtain information at different scales, followed by an SE module to eliminate redundant information, thereby enhancing the network's feature extraction capability, and then a 1×1 standard convolutional block is used to capture additional local

Algorithm 1 Workflow of GLSS2D

Input: Input feature map $X' \in \mathbb{R}^{H\times W\times 2C}$

Output: Fused feature map $X_{\text{Fuse}} \in \mathbb{R}^{H\times W\times 2C}$

Step 1: ***Advanced Feature Decomposition with Local Feature Extraction and Global Dependency Modeling***

Chunk the input feature into X'_1 and X'_2 by channel, $X'_1,\ X'_2 \in R^{H\times W\times C}$

Process X'_1 with parallel convolutions and Process X'_2 with SS2D module

$X_{\text{local}} = \text{Conv}_{1\times1}(X'_1) \oplus \text{Conv}_{3\times3}(X'_1))$
$X_{\text{local}} \in \mathbb{R}^{H\times W\times C}$ captures multilevel local pattern

$X_{\text{global}} = \text{SS2D}(X'_2)$
$X_{\text{global}} \in \mathbb{R}^{H\times W\times C}$ establishes long-range dependency

Step 2: ***Feature FeedForward Enhancement***

Apply shift-window mechanism to X_{local}:
$\tilde{X}_{local} = Shift(X_{local})$
The specific steps are:
(1) *Vertically shift local features to generate $\overline{X}_{local}$;*
(2) *Process $\overline{X}_{local}$ with Conv→BN→ReLU sequence;*
(3) *Shift features horizontally;*
(4) *Reapply Conv→BN→ReLU sequence;*
(5) *Get the final output $\tilde{X}_{local}$ ($\tilde{X}_{local} \in \mathbb{R}^{H\times W\times C}$).*

Refine X_{global} with locality feedback:
$\tilde{X}_{global} = Locality(X_{global})$
The specific steps are:
(1) *Process X_{global} through a 1×1→3×3→1×1 convolutional block to extract refined features $\overline{X}_{global}$;*
(2) *Fuse X_{global} and $\overline{X}_{global}$ via element-wise addition to obtain $\tilde{X}_{global}$;*
(3) *Output the enhanced global feature $\tilde{X}_{global}$ ($\tilde{X}_{global} \in \mathbb{R}^{H\times W\times C}$).*

Step 3: ***Feature Refusion***

Combine enhanced features: $Concat(\tilde{X}_{local}, \tilde{X}_{global})$
$X_{Fuse} = LayerNorm(Concat(\tilde{X}_{local}, \tilde{X}_{global}))$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

spatial information. For the fusion operation of different scale features, the scale-aware module dynamically selects and fuses appropriate scale features and key feature information through self-learning. Specifically, for three different scale feature maps F_A, F_B, and F_C are summed and processed through convolutional and softmax layers to create pixel attention maps $\alpha \in R^{H\times W}$, $\beta \in R^{H\times W}$, and $\gamma \in R^{H\times W}$. Finally, the feature maps obtained from self-learning are weighted and summed to obtain the fused feature maps by elementwise multiplication ($\bigodot$) F_{fusion}

$$F_{\text{fusion}} = \alpha \bigodot F_A + \beta \bigodot F_B + \gamma \bigodot F_C. \tag{10}$$

Finally, the obtained multiscale feature fusion map is concatenated with the standard convolutional branch and the global pooling branch to supplement the context information to obtain the final features. Through the processing of the scale-aware module, SCPP demonstrates excellent noise immunity and effectively mitigates the hole problem.

IV. Experiments

A. Datasets

The effectiveness of our proposed GLVMamba is demonstrated on two datasets: the Vaihingen dataset and the Potsdam dataset (https://www.isprs.org/education/benchmarks.aspx).

1) Potsdam Dataset: The Potsdam dataset contains 38 RSIs, each with a resolution of 6000 × 6000 pixels, covering an area of 3.42 km^2 and containing complex and dense building structures, among others. The dataset comprises five primary foreground categories and one background category. We use three-channel (red, green, and blue) images from multiband true orthophoto (TOP) plots for the experiments. Following [43], [53], [63], and [64], we select 14 images for testing, and the remaining 23 images (excluding images with incorrect annotations 7_10) for training. These images are cropped into small blocks of 1024 × 1024 pixels.

2) Vaihingen Dataset: The Vaihingen dataset is used for semantic annotation tasks in urban environments and includes 33 ultrahigh-resolution TOP plots, averaging 2494 × 2064 pixels. This dataset, like the Potsdam dataset, is used for semantic segmentation and includes six categories. Similarly, we use three-channel (red, green, and blue) images. Similarly, we use 17 images for testing and 16 patches for training. Again, we crop the original images into small chunks of 1024 × 1024 pixels.

B. Implementation Details

1) Training Settings: The experiments are implemented on NVIDIA GTX 3090 GPU using the PyTorch framework. For fast convergence, we use the AdamW optimizer with a base learning rate of 6e^{-4} and dynamically adjust the learning rate through a cosine annealing strategy. For the Potsdam and Vaihingen datasets, the images are randomly cropped into 512 × 512 patches. The data are processed in the same way as [65], and data augmentation techniques such as random scale ([0.5,0.75,1.0,1.25,1.5]), random flip, and random rotation were used in the training process.

2) Loss Function: Due to the category imbalance in the Potsdam and Vaihingen datasets, we use the joint loss of cross-entropy (CE) loss and dice loss. The overall loss is as follows:

$$L = L_{\text{CE}} + L_{\text{Dice}} \tag{11}$$

$$L_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_i \log(\hat{y}_i) \tag{12}$$

$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} \tag{13}$$

where N represents the number of samples, C indicates the number of categories, y_i is the true label of pixel i, $\hat{y}_i$ is the predicted probability that pixel i belongs to the target class, L_{CE} denotes the CE loss, while L_{Dice} represents the dice loss.

3) Evaluation Metrics: We assess the model performance using metrics such as overall accuracy (OA), mean $F1$ (mF1) score, and mean intersection over union (mIoU). They are defined as follows:

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{15}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{16}$$

$$F1 = \frac{2\times \text{Precision}\times \text{Recall}}{\text{Pre} + \text{Recall}} \tag{17}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \tag{18}$$

where TP, TN, FP, and FN denote the true positive, true negative, false positive, and false negative, respectively, mF1 denotes $F1$ score averaged over all the categories, and mIoU denotes the IoU averaged over all the categories.

C. Ablation Study

To assess our proposed method and the two innovative modules, we use the original VSS block without the SCPP module as baseline and perform ablation experiments on the Potsdam dataset. With different hyperparameter choices, we choose the best hyperparameter as a benchmark. The attention drop rate is set to 0, the patch size is set to 4, the d_state value is set to 16, and the random depth rate is set to 0.1. The number of channels in the decoder stages is set to 512, 256, 128, and 64 to match the encoder's channels. The ablation experimental results are presented in Table I.

1) Effectiveness of GLVSS Block: As shown in Table I, baseline represents the results presented with the original VSS block, and there is a large improvement with the GLVSS block compared with the results with the GLVSS block in the second row of the table; when the GLVSS block is added to the network model, mF1, OA, and mIoU improve by 0.33%, 0.25%, and 0.59%, respectively. It can be seen that the "Car" category has the highest improvement with a 0.58% increase in $F1$, and the other four categories "Impervious Surface," "Building," "Low Vegetation," and "Tree" are improved by 0.39%, 0.41%, 0.43%, and 0.26%, respectively.

Furthermore, in the first row of Fig. 6, we can see that the shadows generated by the light pose challenges for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

TABLE I

ABLATION EXPERIMENT OF THE PROPOSED MODULES ON THE POTSDAM DATASET, THE ORIGINAL VSS BLOCK WITHOUT THE SCPP MODULE AS BASELINE (BOLD: BEST RESULTS)

Method	Params(M)	FLOPs(G)	F1(%)					Evaluation index		
			Impervious Surface	Building	Low Vegetation	Tree	Car	mF1(%)	OA(%)	mIoU (%)
Baseline	13.00	5.73	94.13	96.71	87.77	88.73	96.08	92.68	91.58	86.59
Baseline+GLVSS	15.23	6.50	94.52	97.08	87.83	88.97	**96.66**	93.01	91.83	87.18
Baseline+SCPP	21.97	6.91	94.52	97.12	**88.20**	88.99	96.34	93.03	91.94	87.20
Baseline+GLVSS+SCPP (GLVMamba)	24.19	7.67	**94.61**	**97.24**	87.94	**89.23**	96.56	**93.12**	**91.97**	**87.36**

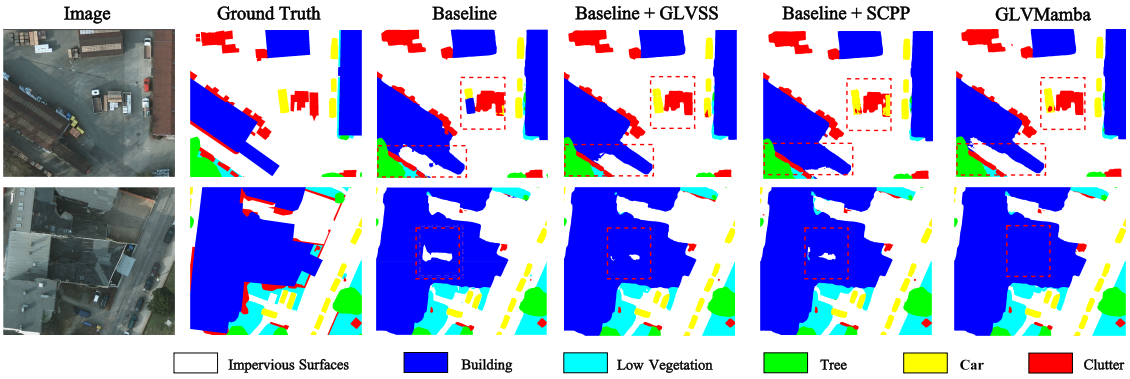


Fig. 6. Ablation experiment visualization of the proposed modules.

model in distinguishing the road from the building, and before adding the GLVSS block, the model recognizes the road as the building and has boundary-blurring problem. With the addition of the GLVSS block, the model effectively improves the ability to recognize objects, correctly identifies the original road part, and effectively improves the target contour and greatly enhances segmentation accuracy for the building boundary. In the middle part, we can also see that before adding the GLVSS block, the right part of the clutter is incorrectly recognized as a car. By incorporating GLVSS, the model almost perfectly segments the car from the clutter. We can also see that under the influence of shadow occlusion, the baseline of applying VSS block has a large holes part, and after our proposed GLVSS block can effectively mitigate the shadow noise interference, reducing the generation of holes, and improving the segmentation performance.

2) Effectiveness of SCPP Module: Table I demonstrates that mF1, OA, and mIoU are improved by 0.35%, 0.36%, and 0.61% after adding the SCPP module to the model, respectively, which exhibits the effectiveness of the SCPP module. The "Low Vegetation" category has the greatest enhancement with a growth rate of 0.43%, followed by the "Building" category which has improved by 0.41%.

As can been seen in the first row of Fig. 6, before the introduction of the SCPP module block, the baseline incorrectly recognizes the part of the car other than the front end as the building class because the rear part of the car is very similar to the texture of the surrounding buildings, which produces a high interclass similarity phenomenon. After incorporating the SCPP module, the model successfully identifies the car part. As can be seen from the second row of Fig. 6, before adding the SCPP module, the texture and color of the middle part of the building are quite different from the other buildings, which makes it difficult for the model to identify and produces a high intraclass variation phenomenon, which leads to the hole phenomenon and misclassification problem. Compared with the ASPP module, with the introduction of the SCPP module, it is able to better adaptively fuse features, leading to a significant reduction in the occurrence of hole within buildings and a substantial enhancement in segmentation performance. From the visualization experiments, it is concluded that the SCPP module mitigates the hole phenomenon and also reduces the false and missed detections of the model.

3) Joint Effectiveness: As shown in Table I, with the combining of both the GLVSS block and the SCPP module, the model's segmentation accuracy increases by 0.39% for mF1, 0.44% for OA, and 0.77% for mIoU. The "Building" category improves the most with 0.53%. This is followed by 0.5% for the "Tree" category. The other three categories, including "Car" and "Impervious Surface," also see improvements of 0.48%, while "Low Vegetation" improved by 0.17%.

As shown in the first row of Fig. 6, GLVmamba accurately identifies cars and clutter, and better distinguishes between buildings and impervious surfaces. As illustrated in the second row of Fig. 6, GLVmamba fully recognizes buildings without generating any holes. This demonstrates that by effectively integrating the GLVSS block and the SCPP module, GLVmamba is able to more accurately identify objects and mitigate the hole phenomenon, significantly enhancing the model's resistance to interference and ultimately yielding more precise segmentation results.

4) Setting the Quantity of GLVSS Blocks in the Decoder: Based on the analysis of the VSS blocks in [29], we investigate the depth of the second stage in the designed GLVSS block decoding block. The experiments are performed on the Potsdam test set with depths of 2, 4, 6, 9, and 12 for the second stage and are evaluated concerning mF1, OA, mIoU, and Params. The specific results are shown in Fig. 7. When

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

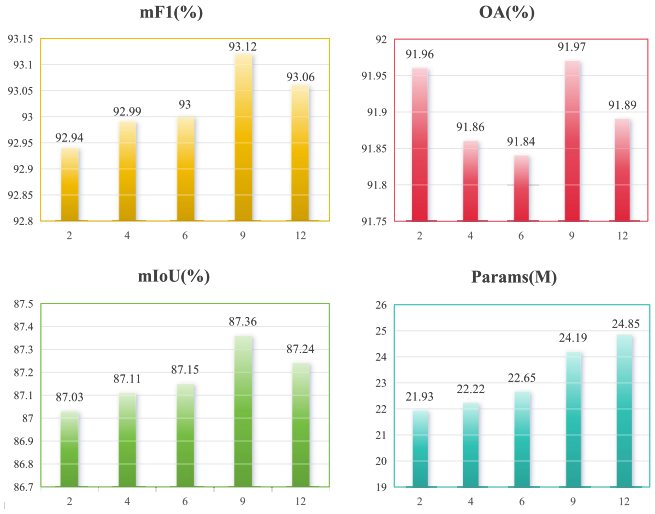


Fig. 7. Bar graph approach illustrates the effect of the number of GLVSS blocks on the segmentation performance. The four plots represent mF1, OA, mIoU, and Parms from the Potsdam test set.

the number of blocks is [2, 9, 2, 2], the model achieves the best segmentation performance, and mF1, OA, and mIoU are 93.12%, 91.97%, and 87.36%, respectively, demonstrating that this configuration provides the optimal balance between model depth and performance. In terms of parameters, it can be observed that the parameters increase with depth. For the depth of the second stage of the decoder, increasing the depth can improve the detail recovery effect and enhance the feature fusion capability. Experiments show that a depth setting of nine blocks can achieve desirable results in processing advanced semantic features while avoiding unnecessary complexity increase. When the depth reaches 12 blocks, the model's performance declines compared with the nine blocks, with mF1, OA, and mIoU decrease by 0.06%, 0.08%, and 0.12%, respectively, likely due to either overfitting or vanishing gradient issues that hinder effective backpropagation. Overall, despite an increase of 1.54M in the number of model parameters in the [2,9,2,2] condition, the mF1, OA, and mIoU reached optimum, and thus [2,9,2,2] is chosen as the experimental block number as it maximizes all the three accuracy metrics while maintaining good generalization ability and avoiding the overfitting observed in deeper models, demonstrating that decoder depth requires careful balancing between semantic extraction capability and model complexity for optimal RSI segmentation.

D. Comparison to the State-of-the-Art Methods

To show the effectiveness of GLVMamba, we analyze GLVMamba in comparison to other methods, including CNN-based methods such as FCN [10], DeepLabV3+ [19], DANet [66], MAResUNet [67], MANet [68], CANet [69], and EIGNet [70], and the SegFormer [71], UPerNet [72], CMTFNet [73], and SCTNet [74] based on hybrid CNNs and transformer models, and also comparing with other methods that use the Mamba structure such as RTMamba [53] and RS^{3}Mamba [31].

1) *FCN:* FCN is capable of classifying input images pixel-by-pixel using convolutional and deconvolutional layers.
2) *DeepLabV3+:* DeepLabV3+ is an enhanced semantic segmentation model that achieves high-precision image segmentation by combining atrous spatial pyramid pooling and encoder–decoder structure.
3) *DANet:* DANet achieves more accurate semantic segmentation by feature enhancement in both the spatial and channel dimensions through the dual-attention mechanism.
4) *MAResUNet:* MAResUNet proposes a linear attention mechanism (LAM) and reconstructed the original U-Net.
5) *MANet:* MANet proposes a network that integrates multiscale awareness through its relational framework, which improves the characterization ability and segmentation accuracy of multiscale information.
6) *CANet:* CANet enhances the integration and combination of features from various levels by introducing the communication mutual attention (CMA) and communication self-attention (CSA) modules, thus improving the accuracy of semantic segmentation.
7) *EIGNet:* EIGNet combines spatial semantic feature aggregation by introducing edge information guidance and orientation convolution to enhance the semantic segmentation accuracy and edge localization accuracy of high-resolution RSIs.
8) *SegFormer:* SegFormer is an efficient semantic segmentation framework that combines a transformer with a lightweight MLP to provide powerful representation capabilities.
9) *UPerNet:* UPerNet is a multitask framework designed for unified perceptual parsing tasks, capable of efficiently segmenting and recognizing multiple visual concepts from images.
10) *CMTFNet:* CMTFNet fuses CNN and multiscale transformer to achieve accurate segmentation of RSIs by combining local information and multiscale global context information.
11) *SCTNet:* SCTNet is a single-branch CNN that incorporates transformer, capturing rich semantic information by introducing transformer branches in training, while retaining only efficient single-branch CNNs in inference for real-time semantic segmentation.
12) *RTMamba:* RTMamba is a novel framework designed for real-time segmentation tasks in RS, combining VSS blocks and inverted triangle pyramid pooling modules to achieve efficient segmentation performance on limited resources platforms while capturing long-range dependencies and boundary awareness.
13) *RS^{3}Mamba:* RS^{3}Mamba is a new type of dual-branch network that combines VSS modules with CNNs, which improves the global context capture and modeling capability of RSIs' semantic segmentation by collaborative completion module.

1) Comparisons on the Potsdam Dataset: We present the quantitative results of other methods on the Potsdam dataset in Table II to further demonstrate the effectiveness of the proposed GLVMamba. As can be seen from the table, our method

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF SEGMENTATION RESULTS ON THE POTSDAM DATASET (BOLD: BEST RESULTS)

Method	F1(%)					Evaluation index		
	Impervious Surface	Building	Low Vegetation	Tree	Car	mF1(%)	OA(%)	mIoU (%)
FCN(2015)	90.84	95.59	84.10	84.75	84.95	88.05	-	79.53
DeepLabV3+(2018)	92.09	95.28	85.55	86.54	94.81	90.86	90.88	84.24
UPerNet(2018)	93.38	96.66	87.89	89.02	96.01	92.59	91.21	86.41
DANet(2019)	90.84	95.59	84.10	84.75	84.95	88.05	-	79.53
SegFormer(2021)	93.45	96.83	88.48	89.98	95.75	92.90	91.48	86.91
MANet(2022)	93.40	96.96	88.32	89.36	96.53	92.90	91.32	86.95
MAResUNet(2022)	90.30	93.81	86.59	88.67	95.93	91.06	-	83.76
CMTFNet(2023)	92.12	96.41	86.43	87.26	92.41	90.93	89.89	83.57
CANet(2023)	93.91	97.22	87.59	88.23	96.07	92.60	91.44	86.48
EIGNet(2023)	93.36	96.86	87.45	89.02	96.51	92.64	91.08	86.53
SCTNet-b(2024)	88.95	94.68	80.71	80.01	92.33	87.34	89.77	83.25
RTMamba-b(2024)	93.91	96.52	88.98	89.98	96.64	92.86	91.56	87.15
GLVMamba(ours)	94.61	97.24	87.94	89.23	96.56	**93.12**	**91.97**	**87.36**

also outperforms the other methods in three metrics, reaching 93.12%, 91.97%, and 87.36%, respectively. Compared with the second-best method, RTMamba, it is important to highlight that RTMamba uses the pretraining weights from VMamba-tiny [29], which was trained on a large number of external datasets, to enhance its effectiveness. In contrast, our model does not rely on the pretraining weights of VMamba-tiny due to modifications in the internal structure of the VSS block. Our approach achieves higher segmentation accuracy than RTMamba across three metrics by solely using its own training data. Specifically, the mF1 score improves by 0.26%, the mIoU increases by 0.15%, and the OA rises by 0.41%. These results demonstrate the potential of the proposed method. On $F1$ in each category, GLVMamba is optimal in "Impervious Surface" and "Building," and the other three categories also show good results.

We visualize a qualitative comparison of GLVMamba with other methods in Fig. 8. In the red boxes in rows 1 and 3, we can see that GLVMamba can achieve better prediction and its performance is better than other methods in dealing with the cases of roads that are difficult to recognize due to being occluded and trees that are hard to distinguish due to similar textures. In addition, in the second row of figure, GLVMamba can effectively address the issue of target object omission and misdetection, realizing more accurate object segmentation and also providing a better target contour. The contribution of the combination of our proposed GLVSS block and SCPP module is effectively verified. However, it can also be seen that in the green box part of the fourth line our model, although it can improve the RSI segmentation in general, it does not perform very well in identifying some dense areas such as cars and clutter with their textures, colors, and their similarities, which we attribute to the fact that it is not sufficient in the local feature extraction of the small target objects, and there is still significant potential for enhancement.

2) Comparisons on the Vaihingen Dataset: The quantitative results of our proposed method with other state-of-the-art methods on the Vaihingen dataset are exhibited in Table III, and the proposed GLVMamba achieves 91.47%, 93.60%, and 84.59% for mF1, OA, and mIoU, respectively. All the three metrics are superior to other methods. Compared with the suboptimal method RTMamba, the mF1 improves by 0.39%, the mIoU also improves by 0.67%, and the OA even improves by 2.3%, and among the $F1$ scores of each category, our proposed GLVMamba achieves the optimal level in Impervious surface, which improves by 3.27% over the suboptimal method.

To evaluate the strengths and weaknesses of other methods compared with our proposed GLVMamba, we visualize the prediction results of several methods in Fig. 9. As can be seen in rows 1, 3, and 4 of the figure, due to noisy images such as lighting, which confuse the semantics of the segmented objects, certain methods such as DANet produce erroneous segmentation phenomena and poor segmentation performance. This is due to these methods do not fully use the global and local context information, which ultimately leads to the misrecognition of objects in the complex region of RSIs. In contrast, GLVMamba is still able to effectively achieve more accurate classification prediction. Meanwhile, in the second row, small objects such as cars are prone to misclassification due to blurred edges and shadow coverage when they are closely connected. The proposed GLVMamba can significantly improve this situation and obtain good segmentation edges. This is consistent with our expectation that the enhancement of segmentation performance can be effectively achieved by combining global and local effective information.

3) Complexity Comparison: We measure the complexity analysis with the number of parameters measured in millions (M) and floating-point operations per second (FLOPS) for each model. All the numbers are obtained on a single NVIDIA 3090 GPU, and the results are shown in Table IV. Most CNN-based models have significantly fewer parameters than transformer-based models. Although the number of parameters is relatively low, the lack of attention to global information makes the CNN-based model inferior, proving that the extraction of global information is crucial. Compared with CMTFNet, it is obvious that the proposed GLVMamba can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

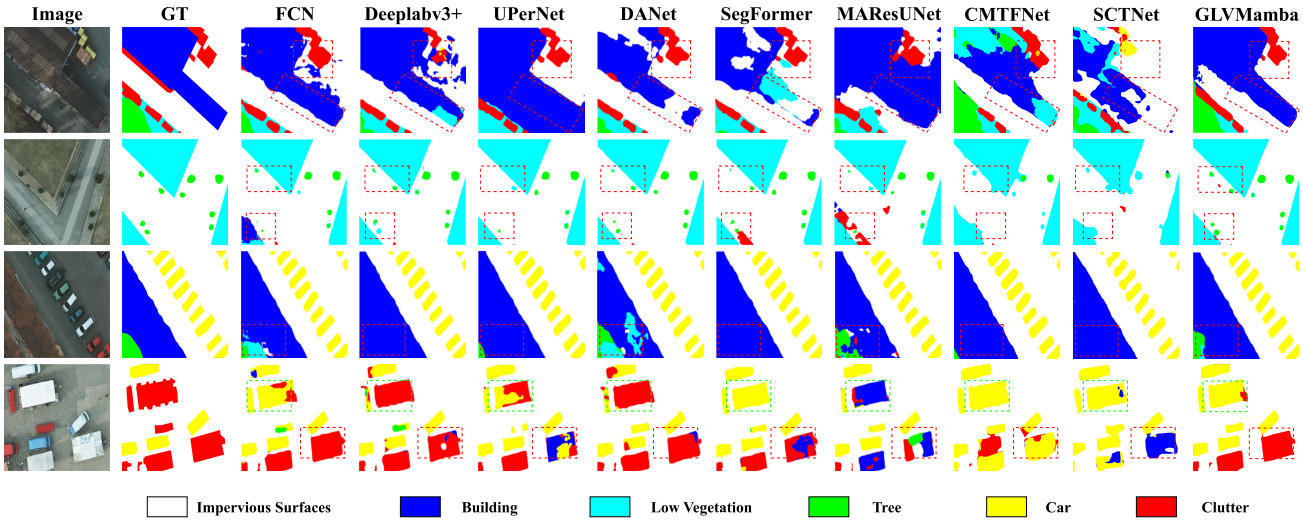


Fig. 8. Comparison of the proposed GLVMamba with other methods on the Potsdam dataset.

TABLE III
COMPARISON OF SEGMENTATION RESULTS ON THE VAIHINGEN DATASET (BOLD: BEST RESULTS)

Method	F1(%)					Evaluation index		
	Impervious Surface	Building	Low Vegetation	Tree	Car	mF1(%)	OA(%)	mIoU (%)
FCN(2015)	89.73	93.17	80.57	88.89	71.55	84.78	-	73.45
DeepLabV3+(2018)	92.38	95.17	84.29	89.52	86.47	89.57	90.56	81.57
UPerNet(2018)	93.30	95.74	84.67	90.91	88.96	90.54	91.15	82.93
DANet(2019)	91.85	94.93	83.81	89.60	80.27	88.09	90.15	79.12
SegFormer(2021)	93.32	95.99	84.58	90.16	89.27	90.67	91.23	83.16
MANet(2022)	93.02	95.47	84.64	89.98	88.95	90.41	90.96	82.71
MAResUNet(2022)	93.27	95.75	85.60	90.79	89.18	90.92	-	83.53
CMTFNet(2023)	90.61	94.21	81.93	87.56	82.77	87.42	88.71	77.95
CANet(2023)	92.49	95.26	83.34	89.18	87.75	89.61	90.30	81.41
EIGNet(2023)	93.17	95.95	85.03	90.01	88.92	90.61	91.21	83.05
RS^{3}Mamba(2024)	86.62	93.83	67.84	83.66	81.97	90.34	-	82.78
RTMamba-b(2024)	93.75	95.30	85.97	90.93	88.70	91.08	91.30	83.92
GLVMamba(ours)	97.02	95.94	85.26	90.41	88.72	**91.47**	**93.60**	**84.59**

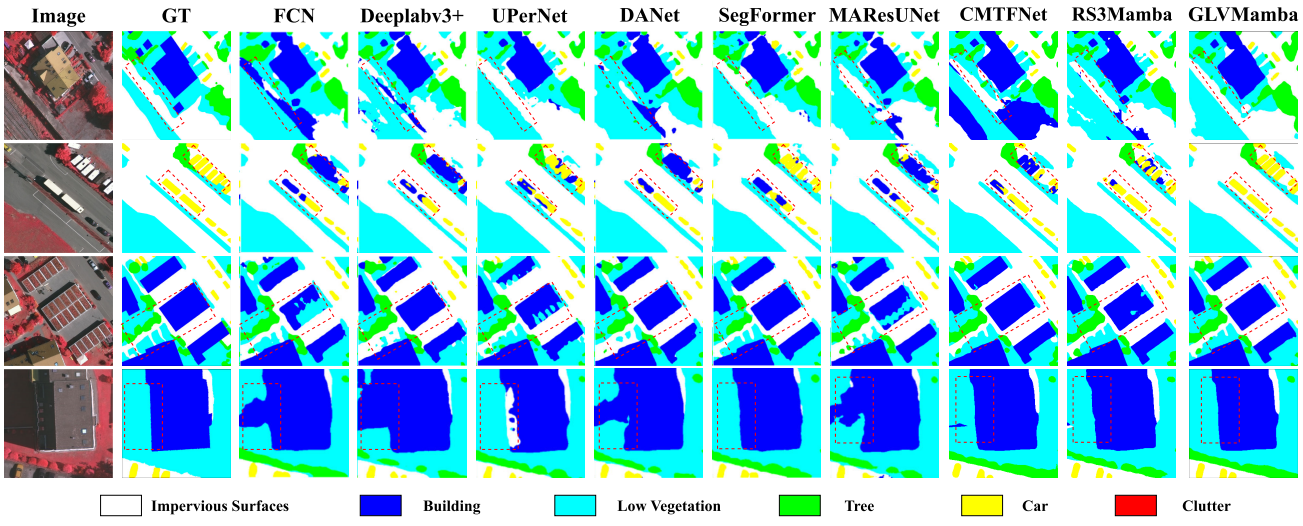


Fig. 9. Comparison of the proposed GLVMamba to other methods on the Vaihingen dataset.

still obtain high segmentation performance with low model scale and computational complexity, which proves that our method is effective in introducing Mamba into RS semantic segmentation tasks. Meanwhile, compared with RS^{3}Mamba,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPLEXITY ANALYSIS OF THE VAIHINGEN DATASET (BOLD: BEST RESULTS, ↓: LOW VALUES ARE BETTER RESULTS, AND ↑: HIGH VALUES ARE BETTER RESULTS)

Type	Method	Params(M) ↓	FLOPs(G) ↓	mIoU(%)↑
CNNs-Based	DeepLabV3+	38.48	20.78	81.57
	DANet	45.36	125.78	79.12
	MANet	35.68	77.62	82.71
	MAResUNet	26.28	17.55	83.53
	EIGNet	34.48	79.70	83.05
Transformer-Based	SegFormer	81.97	25.93	83.16
	UPerNet	102.13	26.03	82.93
	CMTFNet	30.07	32.85	77.95
Mamba-based	RS^{3}Mamba	43.32	31.65	82.78
	GLVMamba(Ours)	**24.19**	**7.67**	**84.59**

the parameter number and FLOPs decreased by 19.13M and 23.98G, respectively, and mIoU also increased by 1.81%, indicating the effectiveness of the improved GLMmamba block.

V. CONCLUSION

In this article, the proposed GLVMamba is an RS semantic segmentation network that combines CNNs and the improved Mamba structures. The GLVSS block is proposed to construct the core decoder for the deficiency of insufficient modeling of neighboring pixel dependencies of SS2D, which enhances the integration of global and local context during feature reconstruction, boosts object perception capabilities of the model, and effectively refines edge contours. In addition, the SCPP module is proposed to fully merge the features from various scales and adaptively fuse and extract the distinguishing features to mitigate the holes and false detections due to the light and other factors. The GLVMamba effectively captures global–local semantic information and multiscale feature through the GLVSS block and the SCPP module, achieving efficient and accurate RS semantic segmentation. Extensive experimental results on the Potsdam and Vaihingen datasets demonstrate that the proposed GLVMamba outperforms the state-of-the-art methods, thereby verifying its effectiveness and superiority.

However, the GLVMamba still has limitations when dealing with dense small objects, primarily due to the lack of modules in our method that addresses boundary blurring and shadow occlusion in these objects. To this end, we plan to further optimize the model's performance in the future by thoroughly investigating boundary detection methods and exploring how to make full use of the limited occlusion pixel information to improve the segmentation accuracy. In addition, we will consider adding validation sets in future work and further evaluate their impact on model performance.

REFERENCES

[1] S. Minaee, Y. Boykov, F. Porikli, A. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3523–3542, Jul. 2021.

[2] Y. Mo, Y. Wu, X. Yang, F. Liu, and Y. Liao, "Review the state-of-the-art technologies of semantic segmentation based on deep learning," *Neurocomputing*, vol. 493, pp. 626–646, Jul. 2022.

[3] Z. Shao, P. Tang, Z. Wang, N. Saleem, S. Yam, and C. Sommai, "BRRNet: A fully convolutional neural network for automatic building extraction from high-resolution remote sensing images," *Remote Sens.*, vol. 12, no. 6, p. 1050, Mar. 2020.

[4] J. Chen, M. Xia, D. Wang, and H. Lin, "Double branch parallel network for segmentation of buildings and waters in remote sensing images," *Remote Sens.*, vol. 15, no. 6, p. 1536, Mar. 2023.

[5] V. Maus, G. Câmara, R. Cartaxo, A. Sanchez, F. M. Ramos, and G. R. de Queiroz, "A time-weighted dynamic time warping method for land-use and land-cover mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3729–3739, Aug. 2016.

[6] F. X. Ru, M. A. Zulkifley, S. R. Abdani, and M. Spraggon, "Forest segmentation with spatial pyramid pooling modules: A surveillance system based on satellite images," *Forests*, vol. 14, no. 2, p. 405, Feb. 2023.

[7] H. F. Tolie, J. Ren, and E. Elyan, "DICAM: Deep inception and channel-wise attention modules for underwater image enhancement," *Neurocomputing*, vol. 584, Jun. 2024, Art. no. 127585.

[8] Y. Li et al., "CBANet: An end-to-end cross-band 2D attention network for hyperspectral change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513011.

[9] L. Sahar, S. Muthukumar, and S. P. French, "Using aerial imagery and GIS in automated building footprint extraction and shape recognition for earthquake risk assessment of urban inventories," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 9, pp. 3511–3520, Sep. 2010.

[10] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[11] Y. Zhang, Z. Qiu, T. Yao, D. Liu, and T. Mei, "Fully convolutional adaptation networks for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6810–6818.

[12] Y. Li, H. Qi, J. Dai, X. Ji, and Y. Wei, "Fully convolutional instance-aware semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2359–2367.

[13] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[14] I. Alonso and A. C. Murillo, "EV-SegNet: Semantic segmentation for event-based cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, p. 0.

[15] A. Abdollahi, B. Pradhan, and A. M. Alamri, "An ensemble architecture of deep convolutional segnet and unet networks for building semantic segmentation from high-resolution aerial images," *Geocarto Int.*, vol. 37, no. 12, pp. 3355–3370, Jun. 2022.

[16] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[17] N. Siddique, S. Paheding, C. P. Elkin, and V. Devabhaktuni, "U-Net and its variants for medical image segmentation: A review of theory and applications," *IEEE Access*, vol. 9, pp. 82031–82057, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

[18] M. Z. Alom, C. Yakopcic, M. Hasan, T. M. Taha, and V. K. Asari, "Recurrent residual U-Net for medical image segmentation," *J. Med. Imag.*, vol. 6, no. 1, Mar. 2019, Art. no. 014006.

[19] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[20] Z. Chang et al., "Crop type identification using high-resolution remote sensing images based on an improved DeepLabV3+ network," *Remote Sens.*, vol. 15, no. 21, p. 5088, Oct. 2023.

[21] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[22] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[23] J. Zhou, M. Hao, D. Zhang, P. Zou, and W. Zhang, "Fusion PSPnet image segmentation based method for multi-focus image fusion," *IEEE Photon. J.*, vol. 11, no. 6, pp. 1–12, Dec. 2019.

[24] X. Chen, D. Li, M. Liu, and J. Jia, "CNN and transformer fusion for remote sensing image semantic segmentation," *Remote Sens.*, vol. 15, no. 18, p. 4455, Sep. 2023.

[25] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[26] X. Xiong, X. Wang, J. Zhang, B. Huang, and R. Du, "TCUNet: A lightweight dual-branch parallel network for sea–land segmentation in remote sensing images," *Remote Sens.*, vol. 15, no. 18, p. 4413, Sep. 2023.

[27] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[28] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[29] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[30] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[31] X. Ma, X. Zhang, and M.-O. Pun, "RS3Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[32] G. Chen, X. Zhang, Q. Wang, F. Dai, Y. Gong, and K. Zhu, "Symmetrical dense-shortcut deep fully convolutional networks for semantic segmentation of very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 5, pp. 1633–1644, May 2018.

[33] Z. Ma, Z. Jiang, and H. Zhang, "Hyperspectral image classification using feature fusion hypergraph convolution neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517314.

[34] B. Dang, Y. Li, Y. Zhang, and J. Ma, "Progressive learning with cross-window consistency for semi-supervised semantic segmentation," *IEEE Trans. Image Process.*, vol. 33, pp. 5219–5231, 2024.

[35] Y. Li, T. Shi, Y. Zhang, and J. Ma, "SPGAN-DA: Semantic-preserved generative adversarial network for domain adaptive remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5406717.

[36] X. Lu, Z. Jiang, and H. Zhang, "Weakly supervised remote sensing image semantic segmentation with pseudo label noise suppression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5406912.

[37] R. Li et al., "DeepUNet: A deep fully convolutional network for pixel-level sea-land segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 3954–3962, Nov. 2018.

[38] Y. Hou, Z. Liu, T. Zhang, and Y. Li, "C-UNet: Complement UNet for remote sensing road extraction," *Sensors*, vol. 21, no. 6, p. 2153, Mar. 2021.

[39] Z. Wang, J. Wang, K. Yang, L. Wang, F. Su, and X. Chen, "Semantic segmentation of high-resolution remote sensing images based on a class feature attention mechanism fused with Deeplabv3+," *Comput. Geosci.*, vol. 158, Jan. 2022, Art. no. 104969.

[40] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lucic, and C. Schmid, "ViViT: A video vision transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6836–6846.

[41] X. Zhou, L. Zhou, S. Gong, S. Zhong, W. Yan, and Y. Huang, "Swin transformer embedding dual-stream for semantic segmentation of remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 175–189, 2024.

[42] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[43] Y. Chen, Y. Wang, S. Xiong, X. Lu, X. X. Zhu, and L. Mou, "Integrating detailed features and global contexts for semantic segmentation in ultrahigh-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703914.

[44] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[45] J. Dong et al., "Dual hyperspectral mamba for efficient spectral compressive imaging," 2024, *arXiv:2406.00449*.

[46] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," 2024, *arXiv:2406.04207*.

[47] W. Zhou, S.-I. Kamata, H. Wang, M.-S. Wong, and H. Hou, "Mamba-in-mamba: Centralized mamba-cross-scan in tokenized mamba model for hyperspectral image classification," *Neurocomputing*, vol. 613, Jan. 2025, Art. no. 128751.

[48] J. Liu et al., "Swin-UMamba: Mamba-based UNet with ImageNet-based pretraining," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Cham, Switzerland: Springer, Feb. 2024, pp. 615–625.

[49] K. S. Sanjid, M. T. Hossain, M. S. S. Junayed, and M. M. Uddin, "Integrating mamba sequence model and hierarchical upsampling network for accurate semantic segmentation of multiple sclerosis legion," 2024, *arXiv:2403.17432*.

[50] Z. Wan et al., "Sigma: Siamese mamba network for multi-modal semantic segmentation," 2024, *arXiv:2404.04256*.

[51] J. Ruan, J. Li, and S. Xiang, "VM-UNet: Vision mamba UNet for medical image segmentation," 2024, *arXiv:2402.02491*.

[52] Q. Zhu et al., "Samba: Semantic segmentation of remotely sensed images with state space model," 2024, *arXiv:2404.01705*.

[53] H. Ding et al., "A novel mamba architecture with a semantic transformer for efficient real-time remote sensing semantic segmentation," *Remote Sens.*, vol. 16, no. 14, p. 2620, Jul. 2024.

[54] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[55] M. Wang, C. Zhu, and X. Chen, "Building change detection in high-resolution remote sensing images based on FPN res-UNet," *J. Jilin Univ., Earth Sci. Edn.*, vol. 51, no. 1, pp. 296–303, Mar. 2021.

[56] Y. Gu, B. Wang, and B. Xu, "A FPN-based framework for vehicle detection in aerial images," in *Proc. 2nd Int. Conf. Video Image Process.*, Dec. 2018, pp. 60–64.

[57] W. Liu, Y. Shu, X. Tang, and J. Liu, "Remote sensing image segmentation using dual attention mechanism deeplabv3+ algorithm," *Tropical Geography*, vol. 40, no. 4, pp. 303–313, Aug. 2020.

[58] Y. Li et al., "MFVNet: A deep adaptive fusion network with multiple field-of-views for remote sensing image semantic segmentation," *Sci. China Inf. Sci.*, vol. 66, no. 4, Apr. 2023, Art. no. 140305.

[59] P. Ma et al., "Multiscale superpixelwise prophet model for noise-robust feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508912.

[60] T. Huang, X. Pei, S. You, F. Wang, C. Qian, and C. Xu, "LocalMamba: Visual state space model with windowed selective scan," 2024, *arXiv:2403.09338*.

[61] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[62] D. Lian, Z. Yu, X. Sun, and S. Gao, "AS-MLP: An axial shifted MLP architecture for vision," 2021, *arXiv:2107.08391*.

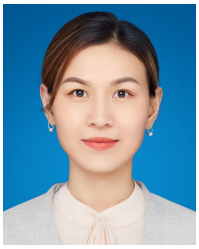
[63] R. Hang, P. Yang, F. Zhou, and Q. Liu, "Multiscale progressive segmentation network for high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412012.

[64] J. Hou, Z. Guo, Y. Wu, W. Diao, and T. Xu, "BSNet: Dynamic hybrid gradient convolution based boundary-sensitive network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604912.

[65] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.

[66] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[67] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[68] P. He et al., "MANet: Multi-scale aware-relation network for semantic segmentation in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624615.
[69] X. Meng, L. Zhu, Y. Han, and H. Zhang, "We need to communicate: Communicating attention network for semantic segmentation of high-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 14, p. 3619, Jul. 2023.
[70] B. Liu, J. Hu, X. Bi, W. Li, and X. Gao, "PGNet: Positioning guidance network for semantic segmentation of very-high-resolution remote sensing images," *Remote Sens.*, vol. 14, no. 17, p. 4219, Aug. 2022.
[71] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.
[72] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.
[73] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.
[74] Z. Xu, D. Wu, C. Yu, X. Chu, N. Sang, and C. Gao, "SCTNet: Single-branch CNN with transformer semantic information for real-time segmentation," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 6, pp. 6378–6386.



Huihui Li (Member, IEEE) received the Ph.D. degree in computer science and engineering from South China University of Technology, Guangzhou, China, in 2019.

She is currently an Associate Professor with the School of Computer Science, Guangdong Polytechnic Normal University, Guangzhou, and a Senior Researcher with the Engineering Technology Research Center of Artificial Intelligence in Traditional Chinese Medicine of Guangdong Province, Guangzhou. Her research interests include remote sensing image analysis, multimedia signal processing, and affective computing.

Dr. Li is a member of the Special Committee on Affective Computing and Understanding of Chinese Society of Image and Graphics.



Huajian Pan received the B.S. degree in data science and big data technology from Guangdong Polytechnic Normal University, Guangzhou, China, in 2023, where he is currently pursuing the master's degree with the School of Computer Science.

His research interests include deep learning and remote sensing image processing.



Xiaoyong Liu received the Ph.D. degree from the National Science Library, Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Professor with the School of Data Science and Engineering, Guangdong Polytechnic Normal University, Guangzhou, China, and the Director of the Big Data Engineering Technology Research Center for the Integration of Production and Education of Guangdong Province, Guangzhou. His research interests include image processing, pattern recognition, and big data analytics.

Dr. Liu is the Vice Chairperson of Guangdong System Engineering Society.



Jinchang Ren (Senior Member, IEEE) received the B.E. degree in computer software, the M.Eng. degree in image processing, and the D.Eng. degree in computer vision from Northwestern Polytechnical University, Xi'an, China, in 1992, 1997, and 2000, respectively, and the Ph.D. degree in electronic imaging and media communication from the University of Bradford, Bradford, U.K., in 2009.

He is currently a Professor with the National Subsea Centre, Robert Gordon University, Aberdeen, U.K., and also a Visiting Professor with Guangdong Polytechnic Normal University, Guangzhou, China. He has published more than 350 articles. His research interests include computer vision and multimedia signal processing, especially on hyperspectral imaging, machine learning, and big data analytics.

Dr. Ren is as an Associate Editor for five international journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *The Journal of the Franklin Institute*, and *Big Data Analytics*.



Zhiguo Du (Senior Member, IEEE) is currently the Chairperson of Guangzhou Engineering Management Institute, Guangzhou, China, and an Associate Professor with South China Agricultural University (SCAU), Guangzhou, where he is also the Director of the Information Network Center. He has published more than 40 papers in journals and conferences and two monograph books. His research interests are network communication, machine learning, artificial intelligence, and big data modeling.

Dr. Du is a Senior Member of China Computer Federation (CCF).



Jingjing Cao received the B.S. degree in geographic information system from Xinyang Normal University, Xinyang, China, in 2011, and the M.S. and Ph.D. degrees in cartography and geographic information system from Sun Yat-sen University, Guangzhou, China, in 2013 and 2019, respectively.

She is currently a Lecturer with the College of Computer Sciences, Guangdong Polytechnic Normal University, Guangzhou. Her research interests include vegetation remote sensing and urban remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:05:05 UTC from IEEE Xplore. Restrictions apply.